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COURSE : COMPUTER VISION COURSE CODE :
ITA 0503

Industry Problem Solving Task

Question 1: Autonomous Vehicle Vision System Selection

An automobile company is developing an autonomous vehicle detection system that must satisfy the following operating constraints:

- If accuracy < 90% → system fails
- If latency > 70 ms → unacceptable
- The vehicle supports moderate computational resources

Three candidate methods were evaluated: Viola-Jones, YOLO, and Deep Learning. Their performance is summarized below.

Method	Accuracy	Latency	Computation	Accuracy ≥ 90%	Latency ≤ 70ms	Computation Check
Viola-Jones	82%	30 ms	Low	❌ Fail	✅ Pass	✅ Pass
YOLO	93%	60 ms	Moderate	✅ Pass	✅ Pass	✅ Pass
Deep Learning	97%	90 ms	High	✅ Pass	❌ Fail	❌ Fail

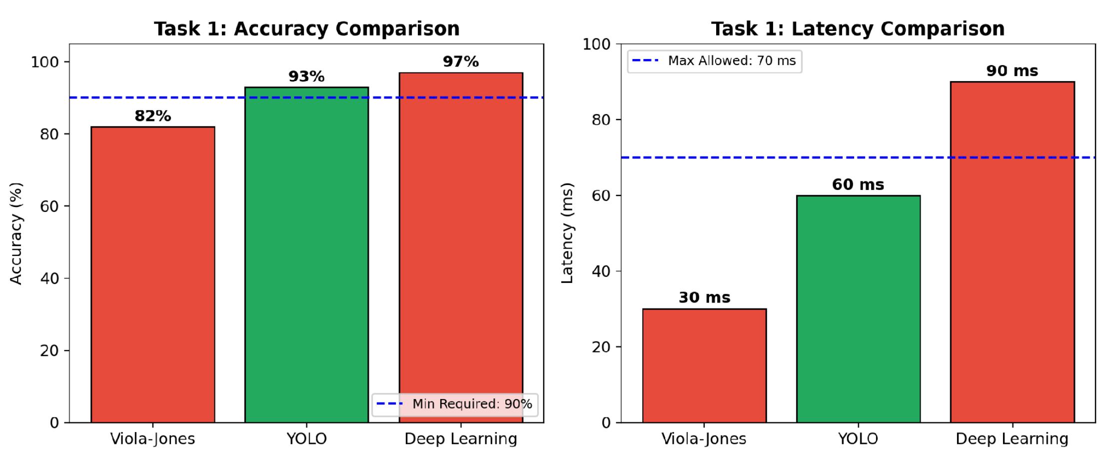


Figure 1: Accuracy and Latency comparison of Viola-Jones, YOLO, and Deep Learning against Task 1 thresholds.

Point-wise Justification

Viola-Jones (82% accuracy, 30 ms latency, Low computation)

- Fastest and lightest of the three methods, but accuracy (82%) is below the required 90% threshold.
- In a driving context, this translates into an unsafe miss-rate for pedestrians, vehicles, or obstacles.

- Rejected on accuracy grounds despite its excellent speed.

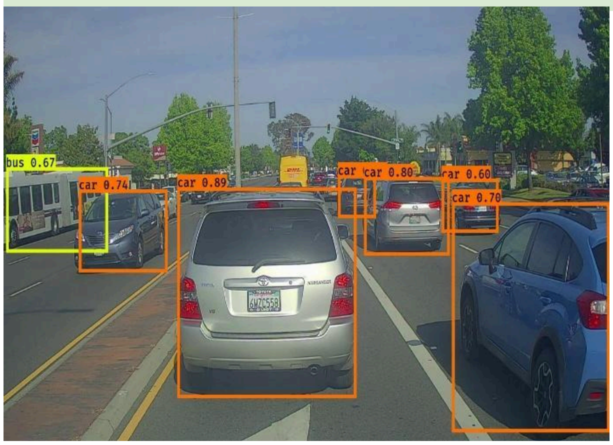
YOLO (93% accuracy, 60 ms latency, Moderate computation)

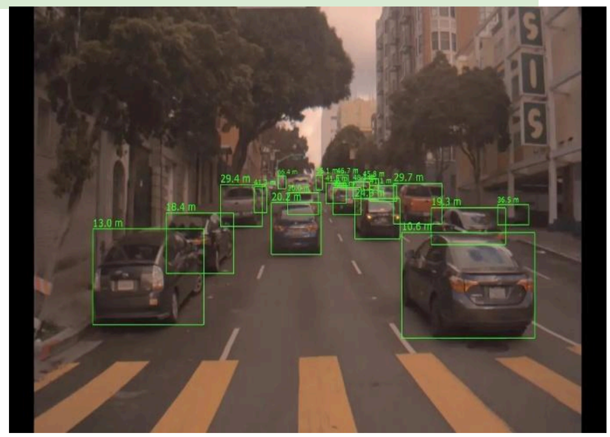
- Accuracy clears the 90% requirement with a comfortable margin.
- Latency (60 ms) stays within the 70 ms real-time limit, leaving a 10 ms buffer.
- Computation load matches the vehicle's moderate hardware exactly.
- Satisfies all three constraints simultaneously.

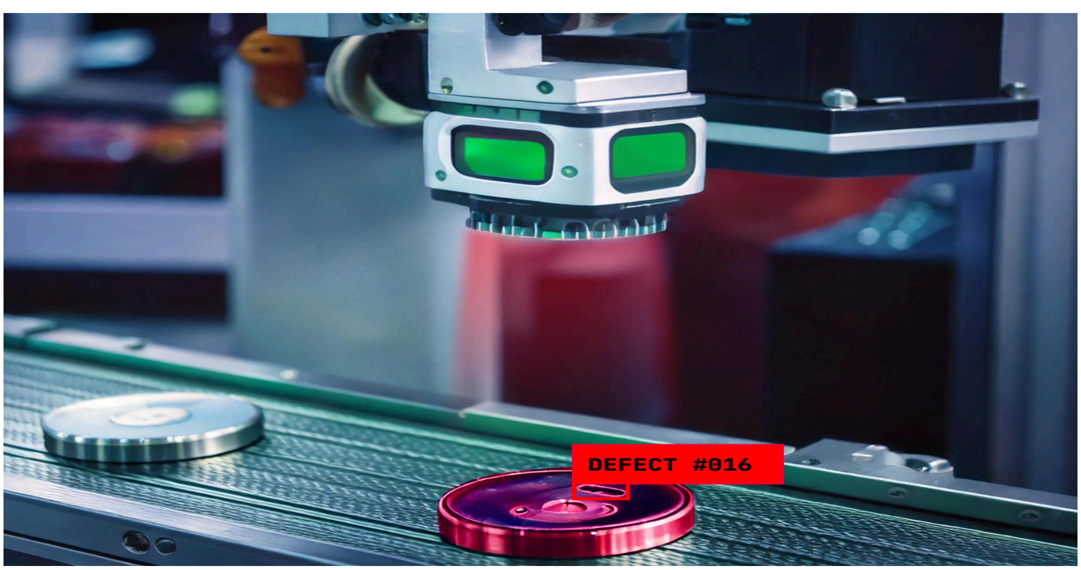
Deep Learning (97% accuracy, 90 ms latency, High computation)

- Highest accuracy of the three methods, but latency (90 ms) exceeds the 70 ms cap by 20 ms.
- Requires high computation — beyond what the vehicle's hardware supports.
- Rejected on two constraints: latency and computation.

Final Decision: YOLO — the only method meeting accuracy, latency, and computation constraints simultaneously.







Question 2: Smart Factory Product Inspection Selection (10 Marks)

A manufacturing company wants to deploy a real-time computer vision system to detect defective products on a production line, subject to the following constraints:

- If accuracy < 95% → defects may be missed
- If processing time > 50 ms → production is affected
- Hardware supports moderate computation

Three candidate methods were evaluated: Traditional Image Analysis, YOLO, and Deep Learning. Their performance is summarized below.

Method	Accuracy	Time	Computation	Accuracy ≥ 95%	Time ≤ 50ms	Computation Check
Traditional Image Analysis	91%	25 ms	Low	❌ Fail	✅ Pass	✅ Pass
YOLO	96%	45 ms	Moderate	✅ Pass	✅ Pass	✅ Pass
Deep Learning	98%	75 ms	High	✅ Pass	❌ Fail	❌ Fail

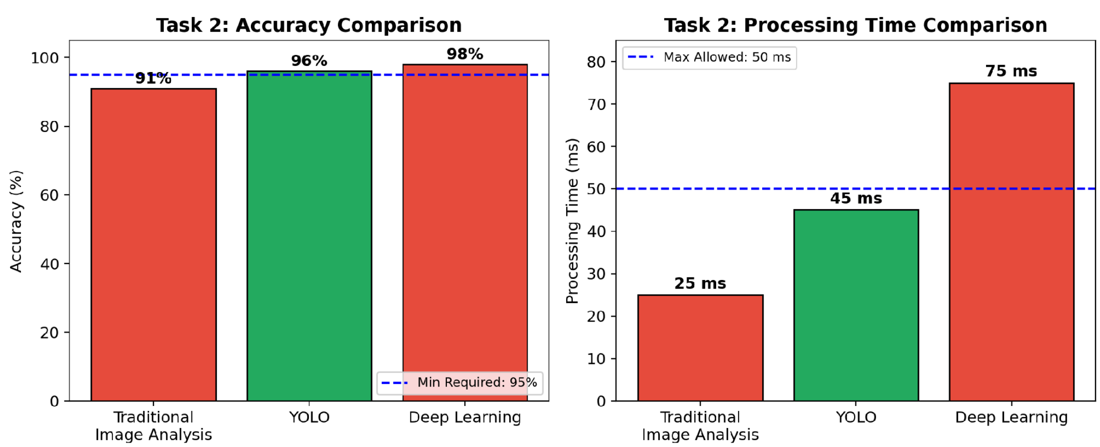


Figure 2: Accuracy and Processing Time comparison of Traditional Image Analysis, YOLO, and Deep Learning against Task 2 thresholds.

Point-wise Justification

Traditional Image Analysis (91% accuracy, 25 ms, Low computation)

- Very fast and lightweight, but accuracy (91%) falls short of the 95% requirement.
- Roughly 1 in 11 defective products could go undetected — a direct quality-control risk.
- Rejected on accuracy grounds despite its speed advantage.

YOLO (96% accuracy, 45 ms, Moderate computation)

- Accuracy (96%) exceeds the 95% threshold, minimizing missed defects.
- Processing time (45 ms) stays under the 50 ms limit, leaving a 5 ms buffer.
- Computation demand matches available factory hardware — no upgrade needed.

- Satisfies all three constraints simultaneously.

Deep Learning (98% accuracy, 75 ms, High computation)

- Highest accuracy of the three, but processing time (75 ms) exceeds the 50 ms limit by 25 ms, bottlenecking the production line.
- Requires high computation, beyond the factory's moderate hardware support.
- Rejected on two constraints: processing time and computation.

Final Decision: YOLO — the only method meeting accuracy, processing time, and computation constraints simultaneously.

Common Takeaway (Applies to Both Tasks)

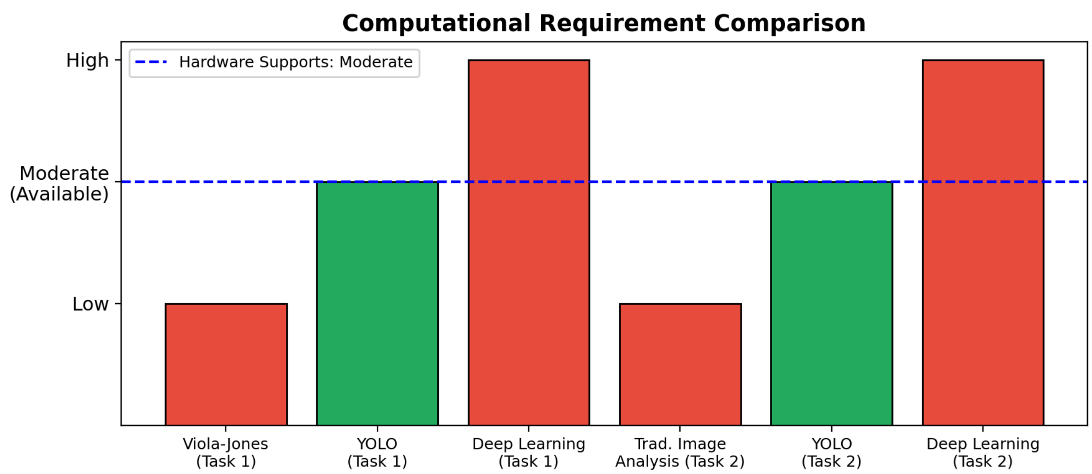


Figure 3: Computational requirement comparison — YOLO is the only method aligned with each system's moderate-hardware constraint in both tasks.

In both cases, the highest-accuracy method (Deep Learning) is disqualified by speed and hardware limitations, while the fastest/lightest method (Viola-Jones / Traditional Image Analysis) is disqualified by insufficient accuracy. YOLO wins both scenarios because it satisfies every constraint simultaneously, not just one metric in isolation — this is the correct basis for method selection in constraint-driven, real-time engineering systems.